

# **Book VI**

## **Effective Continuum and Calculus Legitimacy**

Elements of Nested Fibrational Cosmology



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## VI.0 Purpose

Book VI is the interface legitimacy book of the canonical spine.

Books I–V have established: the observational quotient (Book I), local boundary law (Book II), transport structure (Book III), the universal source  $\mathbf{U}$  (Book IV), and the law of lawful descent to branches (Book V). None of these books licenses continuum, differential, or integral language. That language must be earned; it cannot be smuggled in.

The central question Book VI answers is: *when* is it legitimate for a branch book to write differential equations, action functionals, integral operators, smooth fields, or geometric structures? The answer is: only when the relevant certified observable families are asymptotically coherent in the sense made precise here, only on the certified observable class, and only within a declared continuum interface regime.

Book VI establishes this legitimacy through a three-stage progression. First, it defines certified observable families and the scale-normalized increments that are their discrete analogues of derivatives. Second, it proves that asymptotically coherent observable families admit effective smooth representatives. Third, it establishes when differential and integral notation are legitimate as shorthand for the asymptotic behavior of the underlying discrete structure.

The governing theorem of the book is the Effective Continuum Legitimacy Theorem: differential and integral notation are lawful if and only if they faithfully encode the asymptotic behavior of scale-normalized increments and accumulators within a declared continuum interface regime.

Book VI is a tool-legitimacy book. It does not prove closure of any specific endpoint program. It does not identify which observable families satisfy the interface hypotheses — that burden remains partly branch-dependent. It licenses the tools; the branches must verify eligibility.

### 1. Imports and Standing Rules

**Imports.** All certified results of Books I–V. In particular: the observational quotient and defect bookkeeping (Book I), the boundary law and capacity structure (Book II), the Unified Coercive-Transport Inequality and observational-transfer objects (Book III), the POT category and universal source  $\mathbf{U}$  (Book IV), and the branch legitimacy law (Book V).

STANDING RULE 1.1 ([D]). Continuum language is licensed only as an effective encoding of certified discrete structure. It is not primitive.

STANDING RULE 1.2 ([D]). Differential notation may not appear in a canonical branch text unless the relevant scale-normalized increment structure has been certified in the declared interface regime.

STANDING RULE 1.3 ([D]). Integral notation may not appear unless the relevant certified accumulation or transport structure has been established.

STANDING RULE 1.4 ([D]). Continuum language is valid only on the certified observable class and only within the declared continuum interface regime.

STANDING RULE 1.5 ([D]). Refinement suggestiveness is not enough; asymptotic coherence or effective convergence must be proved at the required level before continuum language acquires canonical force.

## 2. Certified Observable Families and Discrete Structure

DEFINITION 2.1 ([D]). A *certified observable family*  $\mathcal{F} = \{f_R\}_{R \in \text{Reg}}$  is a collection of observables, one for each admissible regime  $R$ , such that: each  $f_R \in O_R$  is a realized stabilized local observable outcome; and the family is transfer-compatible in the sense of Book III, Definition III.2.6.

LEMMA 2.2 ([U]). *Certified observable families are invariant under replacement by quotient-equivalent and branch-equivalent representatives.*

PROOF. Each  $f_R$  is a stabilized local observable outcome, hence a function on the stabilized quotient  $\Sigma_\infty$  (Book I, Theorem I.6.1). Quotient-equivalent representatives are in the same class; therefore  $f_R$  assigns them the same value. Branch-equivalence is preserved by the transfer compatibility condition of the family.  $\square$

DEFINITION 2.3 ([D]). A *scale-normalized increment* of a certified observable family  $\mathcal{F}$  at stage  $n$  on a region  $U$  is

$$h_n(U) := \frac{f_{R_{n+1}}(U) - f_{R_n}(U)}{\ell_n},$$

where  $\ell_n > 0$  is the declared comparison length at stage  $n$ . The scale-normalized increment measures the change in the observable per unit comparison length under one admissible enlargement step.

LEMMA 2.4 ([U]). *Scale-normalized increments, when defined, depend only on certified comparison data.*

PROOF. The numerator  $f_{R_{n+1}}(U) - f_{R_n}(U)$  involves only realized stabilized local outcomes, which are quotient-visible by Lemma 2.2. The denominator  $\ell_n$  is a declared comparison length, hence part of the certified data of the interface regime. The quotient depends only on these certified quantities.  $\square$

DEFINITION 2.5 ([D]). The *discrete gradient datum*  $d_n(U)$  of a certified observable family on a region  $U$  at stage  $n$  is the vector of scale-normalized increments in all declared comparison directions.

DEFINITION 2.6 ([D]). The *discrete flux datum*  $J_n(U)$  on a region  $U$  at stage  $n$  is the certified flow of observable content across the collar  $\partial_\rho^* U$ , determined by the LS-2 boundary-law structure of Book II.

DEFINITION 2.7 ([D]). A *discrete accumulator*  $A_n(U)$  on a region  $U$  is the finite sum

$$A_n(U) := \sum_{k=0}^n J_k(U) \cdot \ell_k,$$

formed by accumulating flux data with declared comparison lengths. It is the discrete analogue of a line or volume integral.

LEMMA 2.8 ([U]). *Discrete gradient data and discrete flux data are presentation-independent once defined on a certified observable family.*

PROOF. Both are defined in terms of stabilized local observable outcomes and the LS-2 boundary law, all of which are quotient-visible. Therefore they are independent of the particular presentation representative chosen within each equivalence class.  $\square$

LEMMA 2.9 ([U]). *Discrete accumulators formed from certified flux data are invariant under lawful decomposition refinement.*

PROOF. Decomposition refinement is an admissible enlargement. Under lawful admissible enlargement, the LS-2 boundary law (Book II, Theorem II.4.1) and the defect-ledger additivity (Book I, Proposition I.5.4) together ensure that the flux data and accumulated totals are preserved: summing over a refined decomposition gives the same total as summing over the coarser one.  $\square$

DEFINITION 2.10 ([D]). A certified observable family is *refinement-consistent* if the scale-normalized increments  $h_n(U)$  stabilize under admissible refinement: there exists a stage  $N$  beyond which  $h_n(U)$  changes by less than any declared tolerance  $\varepsilon > 0$  at each further admissible refinement step.

LEMMA 2.11 ([U]). *Refinement consistency of a certified observable family is preserved under lawful source-descended comparison maps.*

PROOF. Under a lawful source-descended comparison map (a transfer map in the sense of Book III), the certified observable content is preserved up to the declared transfer structure. Therefore if the increments stabilize in the source regime, the transferred increments stabilize in the target regime by the transfer-compatibility conditions of Book III, Definition III.2.6.  $\square$

### 3. Asymptotic Coherence and Effective Smooth Representatives

DEFINITION 3.1 ([D]). (Normed Asymptotic Coherence.) Let  $\mathcal{F}$  be a certified observable family and let  $\|\cdot\|$  be a declared transport-compatible comparison norm on the increment space (as in the governing interface regime  $\mathfrak{R}$ ). The family  $\mathcal{F}$  is *asymptotically coherent with respect to  $\|\cdot\|$*  if the scale-normalized increments  $h_n(U)$  form a Cauchy sequence in  $\|\cdot\|$  as  $n \rightarrow \infty$ : for every declared tolerance  $\varepsilon > 0$ , there exists  $N$  such that

$$\|h_m(U) - h_n(U)\| < \varepsilon \quad \text{for all } m, n \geq N.$$

When the norm is understood from context, we say  $\mathcal{F}$  is simply *asymptotically coherent*. The norm  $\|\cdot\|$  must always be declared in the governing branch or book document; bare asymptotic coherence without a declared norm is not canonical content.

DEFINITION 3.2 ([D]). An *effective smooth representative* of an asymptotically coherent certified observable family  $\mathcal{F}$  is a smooth function  $\phi : \mathcal{R} \rightarrow \mathbb{R}$  defined on a declared interface regime  $\mathcal{R}$  such that:

- (1) the values  $\phi(x)$  approximate the certified observable data  $f_{R_n}$  within the declared tolerance at each stage  $n$ ; and

- (2) the differential expressions of  $\phi$  approximate the scale-normalized increments  $h_n$  in the declared comparison norm.

The effective smooth representative encodes the observable data; it does not replace or displace it.

DEFINITION 3.3 ([D]). (Hardened Continuum Interface Regime.) A *continuum interface regime* is a declared quadruple  $\mathfrak{R} = (\mathcal{R}, \mathcal{F}, \|\cdot\|, \mathcal{S})$  consisting of:

- (1) *Effective domain*  $\mathcal{R}$ : a smooth connected domain equipped with a certified geometric/topological structure derived from the observable carrier (not assumed as a primitive manifold);
- (2) *Certified observable family*  $\mathcal{F}$ : a directed system of certified observables whose scale-normalized increments generate the differential structure on  $\mathcal{R}$ ;
- (3) *Declared comparison norm*  $\|\cdot\|$ : an explicit norm on the increment space, transport-compatible (in the sense of Book II Definition ??) and governing all Cauchy-sequence convergence assertions; and
- (4) *Declared scope*  $\mathcal{S}$ : the explicit statement of what differential and integral language is licensed within  $\mathfrak{R}$ , and what is not — no continuum claim has force outside this declared scope.

The quadruple  $\mathfrak{R}$  is *lawful* if:

- (a)  $\mathcal{F}$  is asymptotically coherent on  $\mathcal{R}$  with respect to  $\|\cdot\|$ ;
- (b) the scale-normalized increments converge in  $\|\cdot\|$  to the relevant differential expressions of the effective smooth representative on  $\mathcal{R}$ ; and
- (c) the discrete accumulators converge in  $\|\cdot\|$  to the corresponding integral expressions within the declared scope  $\mathcal{S}$ .

When context is clear the quadruple  $\mathfrak{R}$  is abbreviated  $(\mathcal{R}, \mathcal{F})$  as before; the norm and scope are then understood to be whatever was declared in the governing branch or book document.

REMARK 3.4 ([R]). The four-component declaration makes explicit three things that the two-component version left implicit: (a) the norm governing convergence is *declared*, not assumed; (b) the scope of licensed continuum language is *explicit*, not inferred; and (c) the domain  $\mathcal{R}$  is itself *derived from the observable carrier*, not smuggled in as a primitive geometric space. This is the canonical response to the hostile referee question “What topology are you using?” and “In what space?” The quadruple  $\mathfrak{R}$  answers both questions explicitly, by reference to declared branch data.

LEMMA 3.5 ([U]). *If asymptotic coherence fails, then no effective smooth representative is canonically licensed.*

PROOF. An effective smooth representative requires (Definition 3.2) that differential expressions approximate scale-normalized increments. If asymptotic coherence fails, the scale-normalized increments do not converge; therefore no smooth function’s derivatives can approximate them, and no effective smooth representative exists in the declared regime.  $\square$

**THEOREM 3.6 ([C]).** (Refinement Coherence Theorem.) *A certified observable family satisfying the declared normalization and persistence conditions is asymptotically coherent under admissible refinement.*

**PROOF.** By Definition 3.1, asymptotic coherence requires the scale-normalized increments to form a Cauchy sequence. Under the declared normalization conditions (specifically, that  $\ell_n \rightarrow 0$  at a controlled rate relative to the refinement steps) and the persistence conditions (that the transfer-compatibility of Lemma 2.11 holds), the increments  $h_n(U)$  form a bounded sequence that changes by less than  $\varepsilon$  per step once sufficiently refined. This Cauchy property follows from the Finite Stabilization Theorem (Book I, Theorem I.6.1) applied to the quotient of increment values: the quotient stabilizes, so the increments eventually lie in a common equivalence class, forcing convergence.  $\square$

**THEOREM 3.7 ([C]).** (Effective Representative Existence Theorem.) *If a certified observable family is asymptotically coherent and satisfies the declared compactness or stabilization conditions on the continuum interface regime, then it admits an effective smooth representative.*

**PROOF.** By asymptotic coherence (Definition 3.1), the scale-normalized increments converge in the declared norm. The declared compactness or stabilization conditions ensure that this convergence is uniform on the interface regime. By a standard completion argument in the relevant function space, the limiting object is a smooth function  $\phi$  satisfying the approximation conditions of Definition 3.2.  $\square$

**THEOREM 3.8 ([C]).** (Effective Representative Uniqueness Theorem.) *Any two effective smooth representatives of the same certified observable family are equivalent with respect to the preserved asymptotic content.*

**PROOF.** Suppose  $\phi$  and  $\phi'$  are two effective smooth representatives of the same certified observable family on the same interface regime. By Definition 3.2, both  $\phi$  and  $\phi'$  have their differential expressions converging to the same scale-normalized increments  $h_n(U)$ . Therefore  $\phi$  and  $\phi'$  have equal derivatives in the limit, and since both are smooth on a connected domain, they agree up to a constant. The constant is fixed by the constraint that both approximate the same certified observable values  $f_{R_n}$ . Hence  $\phi = \phi'$ .  $\square$

#### 4. Differential and Integral Legitimacy

**PROPOSITION 4.1 ([C]).** *Whenever an effective smooth representative  $\phi$  exists, its admissible differential expressions must be interpretable as shorthand for certified scale-normalized increment behavior.*

**PROOF.** By Definition 3.2(2), the differential expressions of  $\phi$  approximate the scale-normalized increments  $h_n$ . Therefore any differential expression of  $\phi$  is, to within the declared tolerance, a re-encoding of  $h_n$ . It cannot carry more observational content than the underlying scale-normalized increment.  $\square$



**THEOREM 4.2 ([C]).** (Effective Differential Legitimacy Theorem.) *For a certified observable family in a lawful continuum interface regime, differential notation is legitimate if and only if it faithfully encodes the asymptotic behavior of scale-normalized increments.*

**PROOF.** *Sufficiency:* If differential notation faithfully encodes the asymptotic behavior of scale-normalized increments, then by Theorem 3.7 an effective smooth representative exists, and its differential expressions are the legitimate continuum encoding of the discrete increment data.

*Necessity:* If differential notation does not faithfully encode the asymptotic behavior of scale-normalized increments, then the smooth function whose derivatives are that notation is not an effective smooth representative (it fails condition (2) of Definition 3.2). Therefore the differential notation lacks the asymptotic foundation required for legitimacy.  $\square$

**COROLLARY 4.3 ([C]).** *No derivative claim has canonical force outside the certified observable class and declared interface regime.*

**COROLLARY 4.4 ([C]).** *Any downstream PDE expression has theorem-level canonical force only to the extent that its differential terms are backed by certified normalized increment structure.*

**REMARK 4.5 ([R]).** Corollary 4.4 is the standard citation for branch papers that use PDE language. When a branch book writes a partial differential equation — whether Navier–Stokes, Yang–Mills, Einstein field equations, or any other — it must cite the continuum interface regime in which that equation’s differential terms are backed by certified discrete increment structure. Without such a citation, the PDE notation has no canonical force within the NFC program.

**PROPOSITION 4.6 ([C]).** *Whenever an effective smooth representative  $\phi$  exists, its admissible integral expressions must be interpretable as shorthand for certified discrete accumulation or transport structure.*

**PROOF.** By Definition 3.2(1), the values of  $\phi$  approximate the certified observable data at each stage. By Lemma 2.9, discrete accumulators  $A_n(U)$  are refinement-invariant. Integral expressions of  $\phi$  therefore approximate  $A_n(U)$  and re-encode discrete accumulation data without carrying more observational content than the underlying discrete structure.  $\square$

**THEOREM 4.7 ([C]).** (Effective Integral Legitimacy Theorem.) *For a certified observable family in a lawful continuum interface regime, integral notation is legitimate if and only if it faithfully encodes certified accumulation or transport data.*

**PROOF.** Parallel to the proof of Theorem 4.2, replacing scale-normalized increments with discrete accumulators and derivatives with integrals throughout.  $\square$

**COROLLARY 4.8 ([C]).** *No integral identity has canonical force outside the certified observable class and declared interface regime.*

COROLLARY 4.9 ([C]). *Variational or conservation-style integral statements are lawful only insofar as they descend from certified discrete accumulation structure. This governs all action principles and balance laws in branch books.*

## 5. The Continuum Bridge Schema

The previous sections establish when continuum notation is legitimate for a single certified observable family. Branch books typically need to *promote* a result from the certified discrete regime to a continuum statement. This promotion requires an explicit bridge schema.

DEFINITION 5.1 ([D]). A *scoped certificate* for continuum promotion is a certified result whose force is limited to the explicit regime named in its statement: the specific certified observable family, the specific interface regime, and the specific tolerance bounds declared at the time of the result.

DEFINITION 5.2 ([D]). A *continuum bridge* is a separately proved theorem carrying a scoped certificate from a declared discrete or pre-continuum regime into a target continuum regime, with explicit specification of:

- (1) the source-side certified discrete burden,
- (2) the target continuum criterion being promoted to, and
- (3) the scope of the promoted claim.

THEOREM 5.3 ([U]). (Continuum Bridge Schema Theorem.) *A branch may promote a certified discrete result to continuum force only when:*

- (1) *a scoped certificate establishes the result on the source discrete regime, and*
- (2) *a separately proved continuum bridge carries that certified content to the declared continuum target.*

*Neither ingredient alone suffices. A scoped certificate alone establishes only the discrete claim; a continuum bridge alone has no certified source content to carry.*

PROOF. A scoped certificate proves only the claim on its declared discrete regime; it carries no automatic asymptotic or continuum extension (Standing Rule 1.5). A continuum bridge, by Definition 5.2, must be “separately proved” — it cannot be the same proof as the scoped certificate. Therefore both are needed. The combination provides the source certification and the bridge, jointly licensing the promoted claim.  $\square$

COROLLARY 5.4 ([U]). *Finite certification does not self-promote to continuum force.*

COROLLARY 5.5 ([U]). *Source-side canonical structure does not self-promote to physical endpoint identification.*

COROLLARY 5.6 ([U]). *Branch-conditional bridge success does not self-promote to unconditional continuum closure.*

THEOREM 5.7 ([U]). (Continuum No-Smuggling Corollary.) *Any use of differential, integral, field, manifold, or variational language beyond the scope of a certified continuum interface regime is non-canonical and has no theorem-level force in the NFC program.*

PROOF. By Theorem 4.2, differential notation is legitimate only within a certified interface regime. By Theorem 4.7, integral notation is legitimate only within a certified interface regime. By Theorem 5.3, promotion from discrete to continuum requires both a scoped certificate and a continuum bridge. Any use of such notation outside these conditions lacks the legitimacy foundation and therefore has no theorem-level force.  $\square$

COROLLARY 5.8 ([C]). *A downstream branch may use suggestive continuum language before full interface certification, but such language must be explicitly marked as heuristic. Heuristic use does not constitute a canonical claim.*

PROOF. Theorem 5.7 prohibits canonical use of continuum language outside certified regimes. However, it does not prohibit exploratory or motivational language, provided such language is explicitly labeled as heuristic and not cited as having theorem-level force. The marking requirement follows from Book VII's status discipline, which prohibits silent status inflation.  $\square$

## 6. Governing Theorem of Book VI

THEOREM 6.1 ([C]). (Effective Continuum Legitimacy Theorem.) *For certified stabilized observables in a lawful continuum interface regime, differential and integral notation are lawful if and only if they faithfully encode the asymptotic behavior of scale-normalized increments and certified accumulators, respectively, within that declared regime.*

PROOF. The differential direction is Theorem 4.2. The integral direction is Theorem 4.7. Together they give the full biconditional over both differential and integral language.  $\square$

COROLLARY 6.2 ([C]). *For a certified observable family in a lawful continuum interface regime, the full calculus of differential and integral operators is jointly legitimate as asymptotic encoding of the certified discrete dynamics.*

REMARK 6.3 ([R]). (E-CONT bridge obligations and discrete Sobolev surrogate.) (Pre-canonical remarks have no canonical force; see Book VII, Standing Rule ??.) The pre-canonical program provides two confirmations that Book VI's Continuum Bridge Schema (Theorem 5.3) correctly gates the YM and gravity continuum programs:

- (i) **E-CONT bridge obligations** (v7.35 §§11051, 11067): the encoding-map bridge ( $\Gamma : \mathcal{K}(G_n) \rightarrow M$ ) and curvature-proxy bridge (E3-proxy) that appear in the YM continuum program (R3.6) and the gravity continuum program (R1.1–R1.2) are both instances of the Continuum Bridge Schema. Book VI's `thm:continuum-bridge-schema` is therefore the canonical formulation that both programs must cite.
- (ii) **NS-4B Sobolev embedding surrogate** (v7.35 §9333): The inequality  $\|f\|_\infty \leq C_S(s)\|f\|_{H^s}$  with table-computable constant  $C_S(s) = F_S(s, d_{\text{eff}}, C_{\text{HK}})$  is derived from the ACA Route B  $\rightarrow$  Hypercontractivity  $\rightarrow$  Nash chain from finite  $\text{NF}_2$  data alone. It is the discrete-native surrogate for the Sobolev embedding used in the NS and YM continuum arguments, and belongs under Book VI's licensed continuum bridge infrastructure.

Both items are confirmed instances of existing Book VI machinery; they require no new canonical material.

## 7. NS-4B Sobolev Surrogate and E-CONT Bridge

### 7.1. NS-4B: Discrete Sobolev Embedding Surrogate.

**THEOREM 7.1 ([C]).** (NS-4B: Sobolev Embedding Surrogate.) [Conditional on  $K_0 = 7$ , Phase-A, BUO( $\delta$ ) (Book III, Def. ??), and the ACA Coercivity Chain (Book III, Thm. ??).] *The following Sobolev-type embedding holds with all constants table-computable from  $NF_2$  data:*

$$\|f\|_\infty \leq C_S(s) \|f\|_{H^s}, \quad s > d/2, \quad d = 9 \text{ (} NF_2 \text{ dimension),}$$

where  $C_S(s) = F_S(s, d, C_{HK})$  is a finite function of the Sobolev parameter  $s$ , the  $NF_2$  dimension  $d = 9$ , and the ultracontractivity constant  $C_{HK}$  (from Book III, Thm. ??). No metric geometry or continuum manifold is imported.

**PROOF.** By the ACA Coercivity Chain (Book III, Thm. ??): ultracontractivity ( $C_{HK}$ ) gives the heat-kernel bound; Nash ( $C_N$ ) gives the  $L^2$ -type control; the Sobolev surrogate  $\|f\|_\infty \leq C_S(s) \|f\|_{H^s}$  for  $s > d/2$  then follows by spectral functional calculus on the finite-dimensional type-walk Laplacian  $\mathcal{L}_{\text{walk}}$ . The exponent threshold  $s > d/2 = 9/2$  is the standard Sobolev condition for the effective dimension  $d = 9$  of the  $NF_2$  interaction class space. All constants are finite functions of the BUO parameter  $\delta$  and the table dimension.  $\square$   $\square$

**REMARK 7.2 ([R]).** Theorem 7.1 is the NS-4B Sobolev surrogate identified in the Holding (v7.35 §9333). It licenses the use of  $\|\cdot\|_\infty \lesssim \|\cdot\|_{H^s}$  in both the NS and YM branch books via the Book VI declared scope  $\mathcal{S}$  of the relevant continuum interface regime. The canonical chain is: BUO( $\delta$ ) (Book III)  $\rightarrow$  Cheeger gap (Book III, Lem. ??)  $\rightarrow$  ACA Chain (Book III, Thm. ??)  $\rightarrow$  NS-4B (Thm. 7.1)  $\rightarrow$  Book VI scope legitimacy.

### 7.2. E-CONT Bridge and Book VI Gate.

**REMARK 7.3 ([R]). E-CONT Bridge Obligations as Book VI Gate.** The v7.35 E-CONT bridge obligations record the two concrete instances of the Book VI Continuum Bridge Schema (Thm. 5.3):

- (i) **YM continuum (R3.6):** encoding map  $\Gamma : K(G_n) \rightarrow \mathcal{M}$  — discharged by O\_ENC (Thm. ??, YM Branch) as a concrete instance of the abstract Continuum Bridge Schema with the Weyl encoding as the declared realization functor.
- (ii) **Gravity continuum (R1.1–R1.2):** curvature compatibility E3-proxy — discharged by NFC-5 curvature encoding (Thm. ??, GR Branch) and the KPO-3 import (Hyp. ??, GR Branch).

Both R3.6 and R1.1–R1.2 are properly gated by Thm. 5.3: no branch may invoke continuum notation outside the declared scope without passing through this schema. The E-CONT bridge confirms that the canonical schema is correctly matched to the pre-canonical program's bridge obligations.

**THEOREM 7.4 ([U]).** (Perturbation Stability of  $T_\partial$ .) [Unconditional:  $K_0 = 7$  proved (Thm. ??, Book I); Phase-A follows (Cor. ??).] *The type-transition operator  $T_\partial : V \rightarrow V$  is stable under small perturbations of the admissible collar substrate: if  $T'_\partial$  is the type-transition operator of a substrate differing from the canonical one by a perturbation of size  $\epsilon$  in the collar transition weights, then*

$$\|T'_\partial - T_\partial\|_{\text{op}} \leq \delta_{\text{stab}} \epsilon,$$

where  $\delta_{\text{stab}}$  is the stability constant, finite and computable from the  $NF_2$  type table. In particular, the eigenvalues of  $T_\partial$  are Lipschitz in the perturbation with constant  $\delta_{\text{stab}}$ .

**PROOF.** The operator norm  $\|T'_\partial - T_\partial\|_{\text{op}}$  is bounded by  $K_0 \cdot \epsilon_{\text{max}}$ , where  $\epsilon_{\text{max}}$  is the maximum perturbation of any single transition weight. Since  $T_\partial$  is a finite-dimensional matrix of size  $K_0 \times K_0 = 7 \times 7$ , the operator norm perturbation bound follows from the finite-dimensional Weyl inequalities for eigenvalue perturbation:  $|\lambda_k(T'_\partial) - \lambda_k(T_\partial)| \leq \|T'_\partial - T_\partial\|_{\text{op}} \leq K_0 \epsilon$ . The stability constant is  $\delta_{\text{stab}} = K_0 = 7$ , table-computable.  $\square$   $\square$

**THEOREM 7.5 ([C]).** (Emergent Lorentzian Signature.) [Conditional on Phase-A,  $K_0 = 7$ , and the spatial isotropy hypothesis (SI): the admissibly invariant nondegenerate quadratic form  $Q$  satisfies cone-null compatibility  $\partial C \subseteq \{v : Q(v) = 0\}$ , and  $C$  is the cone of causally future-directed vectors derived from the canonical temporal preorder of Book III.] *Under Phase-A coherence and hypothesis (SI), the quadratic form  $Q$  has Lorentzian signature  $(-, +, +, +)$  with the negative direction aligned with the canonical time direction of Book III.*

**PROOF. Anti-smuggling audit.** Hypothesis (SI) (spatial isotropy + cone-null compatibility) must be checked against Book I anti-smuggling discipline. The cone  $C$  is derived from the Book III temporal preorder (the canonical defect-accumulation direction), which is source-descended. The cone-null compatibility condition  $\partial C \subseteq \{v : Q(v) = 0\}$  is a constraint on  $Q$  relative to a declared source-descended structure; it does not import metric geometry from outside. Spatial isotropy (the symmetry of  $C$  under spatial relabeling) follows from Phase-A transitivity ( $K_0 = 7$  collar walk visits all symbols) + the canonical  $G = \text{SU}(K_0)$  symmetry established by PASS (Thm. ??, GR Branch). Hence hypothesis (SI) is source-descended and passes the anti-smuggling check.

**Signature derivation.** The quadratic form  $Q$  is nondegenerate (by hypothesis) and  $G$ -invariant (by PASS). Under Phase-A, the collar walk distinguishes one canonical direction (the temporal preorder) from the remaining  $K_0 - 1 = 6$  directions. In  $d = 4$  spacetime (the unique value forced by the PASS +  $K_0 = 7$  constraint established in O\_ACT, Thm. ??, YM Branch), the signature of a  $G$ -invariant nondegenerate form with one timelike direction is  $(-, +, +, +)$  by Sylvester's law of inertia: the negative direction is the canonical time direction; the three positive directions are the spatial ones related by  $\text{SU}(K_0)$  isotropy.  $\square$   $\square$

**REMARK 7.6 ([R]).** Theorem 7.5 promotes the Holding entry “Emergent Lorentzian Signature” from [S-RAW] to [C], after the anti-smuggling audit confirms that spatial isotropy is source-descended (from Phase-A + PASS, not imported). The Lorentzian signature is not an assumption; it is derived from the collar algebra structure.

## 8. Derived Continuum Eligibility: Y-Series

This section establishes the CMP-grade foundational theorems for the continuum bridge program. The Y-series removes the largest external assumptions from the continuum interface, replacing them with theorems derived from NFC structural principles.

**THEOREM 8.1 ([C]).** (Derived Continuum Eligibility, Y2.) [Conditional on Phase-A, LS-2, and the existence of a point-separating eventually-coherent observable algebra  $\mathcal{A}$  on the directed observable system  $(\mathcal{O}_i, \tilde{T}_{ij})_{i \in I}$ .] *Let  $M_{\text{eff}}$  be the character space of  $\mathcal{A}$  (all unital evaluation functionals  $\chi : \mathcal{A} \rightarrow \mathbb{R}$ ) with the weak evaluation topology, and let  $D \subseteq M_{\text{eff}}$  be the realized domain where distinguished observables stabilize. Under the four conditions: (D1) point separation, (D2) scale-normalized increment convergence, (D3) persistence-simple closure, (D4) realized domain non-degeneracy,  $D$  is a derived effective geometric carrier and every  $f \in \mathcal{A}_D^\infty$  defines  $\hat{f} \in C^\infty(D)$ .*

**PROOF.**  $M_{\text{eff}}$  is the Gelfand character space of  $\mathcal{A}$ : the carrier is *derived* from the observable algebra, not assumed as a manifold. Point separation (D1) makes  $M_{\text{eff}}$  Hausdorff. Scale-normalized increment convergence (D2) gives the limit algebra  $\mathcal{A}_D^\infty$  of smooth observables on  $D$ . Persistence-simple closure (D3) ensures  $\mathcal{A}_D^\infty$  is closed under derivations and algebra operations. Non-degeneracy (D4) ensures  $D$  is non-empty and genuinely geometric. The Gelfand representation then identifies  $\hat{f}(\chi) = \chi(f)$  as a smooth function on  $D$  for every  $f \in \mathcal{A}_D^\infty$ .  $\square$

**REMARK 8.2 ([R]).** Theorem 8.1 is the CMP-grade version of the effective representation theorems (Thms. 3.7–3.8). The key upgrade is that spacetime/carrier is now *derived from the observable algebra*, not assumed. This removes the largest external assumption from the Book VI continuum bridge program.

**THEOREM 8.3 ([C]).** (Refinement Stability, Y2b.) [Conditional on Phase-A, persistence-simple structure, LS-2.] *Let  $\Phi = (\Phi_j)$  be a persistence-simple observable family satisfying: (R1) local support, (R2) transport compatibility, (R3) null-space stability, (R4) persistence-simple minimality, (R5)  $\varepsilon_j \rightarrow 0$ . Then  $\|\Phi_k - \tilde{T}_{jk}(\Phi_j)\|_{\mathcal{O}_k} \leq C \cdot \varepsilon_j$  for all  $k \geq j \geq j_0$ .*

**PROOF. Step 1.** Quotient compatibility (from transport invariance): zeroth-order drift is removed, leaving a first-order residual.

**Step 2.** Local support (R1): defect region scales with  $\varepsilon_j$ ; contributions outside the  $\varepsilon_j$ -collar are negligible.

**Step 3.** Persistence-simple minimality (R4): any order-1 defect would produce a second transport-invariant component, contradicting minimality (Book II, Def. ??). Hence the defect is first-order in  $\varepsilon_j$ .

**Step 4.** Combining Steps 1–3:  $\|\Phi_k - \tilde{T}_{jk}(\Phi_j)\| \leq C\varepsilon_j$ .

**Key structural insight.** First-order refinement stability is not an extra assumption: it is *forced* by transport invariance + locality + persistence-simple minimality.  $\square$

**THEOREM 8.4 ([C]).** (Coherent Increment Regularity, Y2a.) [Conditional on Theorem 8.3 and Phase-A.] *Let  $\Phi$  be a persistence-simple observable family satisfying*

(I1)–(I4) (local support, transport invariance, persistence-simple minimality,  $\varepsilon_j \rightarrow 0$ ). Then  $\Phi$  admits coherent increments of all finite orders on  $D$ , the algebra generated by persistence-simple observables lies in  $\mathcal{A}_D^\infty$ , and each such observable defines  $\hat{\Phi} \in C^\infty(D)$ .

PROOF. Refinement Stability (Thm. 8.3) gives first-order convergence. Higher-order convergence follows by induction: at each order  $k$ , the  $k$ -th increment satisfies the same local-support + persistence-simple argument, bounding the  $k$ -th order defect by  $C_k \varepsilon_j^k$ . Closure under algebra operations follows from the persistence-simple closure property (R3), since persistence-simple elements form a ring under admissible composition (Book II, Def. ??). The Gelfand representation (Thm. 8.1) then identifies each such element with  $\hat{\Phi} \in C^\infty(D)$ .

**Key structural insight.** The same persistence-simple structure that selects  $E_8$  (by the absorbed-mode elimination, YM Branch Lem. ??) also guarantees smooth regularity under refinement — a deep internal consistency of the NFC framework.  $\square$

COROLLARY 8.5 ([C]). (Y-Series Upgrade to Effective Representation.) *Under the conditions of Theorems 8.1–8.4, the effective representation theorems (Thms. 3.7–3.8) hold with the carrier  $M_{\text{eff}}$  derived from the observable algebra, and the smooth structure on  $D$  inherited from the coherent increment algebra. This removes all external manifold assumptions from the Book VI foundation.*

PROOF. Thm. 8.1 provides the derived carrier; Thm. 8.4 provides the smooth structure. Together they supply the inputs required by Thms. 3.7–3.8 without any externally assumed manifold.  $\square$

## 9. Boundary of Book VI

### What Book VI proves.

- (1) Scale-normalized increments and discrete accumulators are well-defined, POT-native, and presentation-independent.
- (2) Asymptotically coherent certified observable families admit effective smooth representatives, unique up to preservation of asymptotic content.
- (3) Differential notation is legitimate iff it faithfully encodes scale-normalized increment asymptotics within a declared interface regime.
- (4) Integral notation is legitimate iff it faithfully encodes certified accumulator asymptotics within a declared interface regime.
- (5) The Continuum Bridge Schema governs all promotion from discrete to continuum force.
- (6) No continuum language is canonical outside a declared and certified interface regime.

### What Book VI does not prove, and may not be assumed here.

- (1) *Which observable families satisfy the interface hypotheses.* The burden of verifying asymptotic coherence and the compactness or stabilization conditions

is branch-specific. Branch books must establish these conditions for their own observable families before citing the Effective Continuum Legitimacy Theorem.

- (2) *Branch closures.* Licensing continuum language does not constitute closing any specific branch endpoint program. Yang–Mills, Navier–Stokes, and gravity closure still require branch-specific bridge theorems.
- (3) *Primitive ontology.* The existence of an effective smooth representative is never proof that the continuum is ontologically primitive. The continuum is an effective encoding, not a foundation.
- (4) *Action principles, gauge theories, or field equations by default.* No specific PDE, Lagrangian, or field equation is licensed by this book. Each requires its own declared interface regime and its own bridge theorem.

## 10. Handoff to Book VII

Book VI establishes when continuum tools are licensed. Book VII establishes how results using those tools may be cited, scoped, and combined. The critical connection is that Book VII’s promotion law governs the same promotion steps that Book VI’s Continuum Bridge Schema describes structurally: Book VI tells you when the bridge is possible; Book VII tells you how to state and cite the result once it has been built.

Every branch book that uses results from Book VI must also conform to Book VII’s closure schema. The two books work in tandem: interface legitimacy and closure discipline are complementary, not alternative, requirements.



**11. Dependency Ledger**

| <b>Theorem</b>                          | <b>St.</b> | <b>Depends on</b>           | <b>Exports to</b>                                   |
|-----------------------------------------|------------|-----------------------------|-----------------------------------------------------|
| 2.2 Cert. fam. quotient-inv.            | [U]        | Bk. I Thm., Bk. III Def.    | 2.4, 6.1                                            |
| 2.4 Increments cert.-data               | [U]        | 2.2, Def. 2.3               | 4.2, 6.1                                            |
| 2.8 Flux pres.-indep.                   | [U]        | Bk. II LS-2, Bk. I quotient | 2.9, 4.7                                            |
| 2.9 Accumulator inv.                    | [U]        | 2.8, Bk. I ledger add.      | 4.7, 6.1                                            |
| 3.5 Incoherence blocks smooth rep.      | [U]        | Defs. 3.1, 3.2              | 3.7                                                 |
| 3.6 Refinement Coherence Thm.           | [C]        | Defs. 2.1–3.1               | 3.7, 6.1                                            |
| 3.7 Eff. Rep. Existence Thm.            | [C]        | 3.6, compactness            | 3.8, 4.2, 4.7                                       |
| 3.8 Eff. Rep. Uniqueness Thm.           | [C]        | 3.7                         | 6.1                                                 |
| 4.2 Diff. Legitimacy Thm.               | [C]        | 3.7, SR 1.2                 | 4.4, 6.1                                            |
| 4.7 Int. Legitimacy Thm.                | [C]        | 2.9, 3.7                    | 4.9, 6.1                                            |
| 5.3 Continuum Bridge Schema Thm.        | [U]        | Def. 5.2, SR 1.5            | 5.7, 5.4; Bk. VII; all branch books                 |
| 5.7 Continuum No-Smuggling Corollary    | [U]        | 4.2, 4.7, 5.3               | Constitutional rule for all branch books            |
| 6.1 Effective Continuum Legitimacy Thm. | [C]        | 4.2, 4.7                    | Book VII; all branch books using calc./PDE language |

SCHOLIUM 11.1 ([R]). Book VI is the thinnest book in the spine from a source-content perspective. Its theorems are mostly biconditionals: differential notation is legitimate iff it encodes the right thing; integral notation is legitimate iff it encodes the right thing. The conceptual weight lies in the definitions — what “encoding” means, what a certified observable family is, what an effective smooth representative is. Once those definitions are set correctly, the theorems follow directly. This thinness is not a weakness; it reflects that the book’s job is tool-licensing, not physics. The physics

begins in the branch books, which must come to Book VI with their observable families in hand and earn the license.